

Lab 3 — Architectural Style Selection & Justification

Self-Service Coffee Kiosk System

Architecture Selection

We chose **Layered Architecture** for the Self-Service Coffee Kiosk System.

Architectural Choice & Reasons

1. Clear separation of concerns: the touchscreen interface, Order Manager, payment processing, and menu/pricing data can be organized into distinct layers. This makes the kiosk easier to understand, test, maintain, and modify.

2. Appropriate for a focused kiosk: the system has a small, well-defined set of functions—three coffee types, two sizes, credit-card payment, receipt printing, and menu/pricing storage. It does not require the operational complexity of independently deployed microservices.

Security Advantage

The layered structure separates the customer-facing interface from payment and data handling. Payment requests can be validated and controlled by the Payment Service rather than exposing payment-processing logic directly to the touchscreen layer. This reduces unnecessary access between components.

Performance Benefit

The architecture keeps local menu/pricing access and order processing within a simple component flow, reducing unnecessary network hops and coordination overhead. This supports quick touch-screen interactions and fast order processing in a busy café.

Identified Components

Touchscreen UI; Order Manager; Payment Service; Menu & Pricing DB; Receipt Printer Service. The diagram shows four key interfaces: order requests, payment processing, menu/pricing queries, and receipt printing.